

Devadarrsha P D

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Executive Summary

I am a Computer Science Engineering student experienced in designing and implementing intelligent, data driven solutions across predictive modeling, machine learning, blockchain, and automation. Proficient in Python, C, C++, SQL, and modern development frameworks, with a track record of building end-to-end systems from AI driven predictive models to Web3-integrated platforms. Passionate about leveraging emerging technologies to create real-world impact, optimize user experiences, and drive digital transformation. Eager to apply technical expertise and collaborative skills in a fast-paced, technology driven environment to drive measurable impact.

Education

Vellore Institute of Technology, Chennai

Bachelor of Technology in Computer Science Engineering

Chennai, India

2025 – 2029

Computer Software College

Advanced diploma in Artificial Intelligence and Data Science.

Final Grade: [A]

Tambaram, Chennai

June 2025 – October 2025

Alwin Memorial Public School

10th Grade | Score: 470/500

12th Grade | Score: 426/500

Selayur, Chennai

2022-2023

2024-2025

Projects

Aircraft Engine Health Monitoring & Predictive Maintenance | Machine Learning, Aviation Analytics

- Built a Remaining Useful Life (RUL) prediction system for turbofan engines using the NASA C-MAPSS dataset, analyzing telemetry such as temperature, pressure, vibration, and fuel flow to detect abnormal operating conditions.
- Designed a hybrid architecture combining an **LSTM** network for engine health regression (RUL/SOH estimation) with **Random Forest** and **Gradient Boosting** classifiers run in parallel for anomaly and fault classification.
- Implemented a threshold-based alarm system to flag degrading engine health and mapped fault alarms to physical failure modes and subsystems using sensor cluster analysis.
- Integrated **SHAP**-based explainability to support natural-language querying of model predictions, along with visualizations of engine degradation trends over time.

GreenLedger – AI Energy & Carbon Optimization Platform | Full-Stack, Machine Learning, Web3

- Developed a full-stack platform that estimates real-time power consumption from Windows system telemetry using an **XGBoost** regression model, then converts predicted energy usage into a carbon footprint estimate.
- Built user-approved optimization recommendations with before/after impact comparison, rewarding verified energy savings through a **Green Credits** system.
- Designed and deployed a **Web3 marketplace** for NFT-based sustainability badges (ERC-721/1155) on the Ethereum Sepolia testnet using Solidity and MetaMask wallet integration.
- Implemented the stack with a Next.js/TypeScript/Tailwind frontend, FastAPI backend, and a Python-based Windows telemetry agent, deployed via Vercel.

Prism - Power Regulation and Intelligent Saving Mechanism | IoT, Embedded Systems

- Developed an intelligent IoT-based smart energy management system using Arduino Uno, PIR Sensor, Ultrasonic Sensor, LDR, PWM, and Python to optimize energy consumption in residential environments.
- Implemented sensor fusion logic by combining PIR and Ultrasonic sensors to improve occupancy detection accuracy and eliminate false triggers.
- Designed a **weighted utility algorithm** for adaptive LED brightness control based on occupancy, ambient light intensity, and energy availability.
- Integrated PySerial-based communication and cloud telemetry for real-time monitoring, data logging, and remote

visualization through IoT dashboards.

- *Achieved approximately **67.7% energy savings** compared to traditional lighting systems through daylight harvesting and intelligent dimming strategies. give me same*

Skills

- **Programming Languages** : Python, C, C++, SQL
- **Machine Learning** : Scikit-learn, TensorFlow, Pandas, NumPy, NLP, SHAP based explainability, Gradient Boosting
- **Web Development** : Next.js, TypeScript, Tailwind CSS, FastAPI
- **Blockchain** : Solidity, ERC-721/1155, MetaMask, IPFS, Ethereum testnets (Sepolia)
- **Tools** : Git, GitHub, VS Code
- **Fundamentals** : Data Structures & Algorithms, Object-Oriented Programming, Probability & Statistics, Artificial Intelligence and Machine Learning Fundamentals
- **Soft skills** : Leadership skills, Communication, Critical Thinking

Achievements

- *Won overall 2nd Place at Recursion 2.0 (24-hour hackathon organized by Microsoft Innovation Club, VIT Chennai) for GreenLedger, an AI-powered energy and carbon optimization platform with Web3 rewards integration.*

Certifications

- **Version Control | Meta (via Coursera)**
 - *Developed skills in Git, GitHub, Version Control, Branching & Merging, Pull Requests, and Collaborative Development through Meta's industry recognized certification program.*
- **Google AI Essentials | Google (via Coursera)**
 - *Developed skills in Generative AI, Prompt Engineering, AI Productivity Tools, and AI-driven Problem Solving through Google's industry recognized certification program.*